

LMS Bridge: a free AI-guided learning tool TWAS can offer its members

Concept note for The World Academy of Sciences (TWAS) / Young Affiliates Network (TYAN) · Hasan Aljabbouli, New York University

In one line

LMS Bridge is a free, open-source platform that turns assessment results into just-in-time, AI-guided tutoring for foundational STEM concepts. It is already built and working. This note proposes that **TWAS adopt it as a member benefit** — try it, then offer it to members, optionally hosted by TWAS and presented under a TWAS name. **No funding is requested.**

The problem it solves

In cumulative STEM subjects, early misconceptions (binary representation, inheritance, probability) compound and quietly sink students weeks later. Large classes make individual, concept-level feedback impossible to deliver in time. This gap is sharpest where teaching resources are thin — exactly the communities TWAS serves.

What LMS Bridge does

- Detects likely misconceptions from quiz and assignment results, at the concept level.
- Generates short, pedagogically constrained tutoring — retrieval practice, Socratic scaffolding, mastery progression. It diagnoses and guides; it never hands over answers.
- Works two ways: **plug into an existing LMS** via LTI 1.3 (Canvas, Moodle, Brightspace, Blackboard), or run **Safe**, its standalone mini-LMS that needs no LMS and no IT — an instructor shares a join code and starts in minutes.
- Privacy-first: self-hostable, bring-your-own AI key, no telemetry, FERPA-aware.

Why this fits TWAS

- **Mission:** strengthens science capacity in the developing world; supports SDG 4; a reusable public good members can share South–South.
- **A benefit, not a cost:** free and open-source (AGPL-3.0). No licensing fee, no procurement hurdle.
- **Brandable:** it can run under a TWAS name and logo (e.g. "TWAS Learning") on a TWAS subdomain or a special link, while staying the same upstream open-source project.
- **Full control:** TWAS hosts its own instance and owns its data; nothing is locked to a vendor.

How TWAS could offer it to members

- **Simplest:** TWAS hosts one branded instance; any member instructor signs in and shares a join code with students. Zero IT for the member.
- **Self-host:** a member institution runs its own copy and plugs it into its LMS via LTI, for full data sovereignty.
- **List & endorse:** TWAS simply lists it as a recommended member resource and points to it.

The one honest cost

The software is free. Running a shared instance costs only a small server plus AI inference for the tutoring calls. This stays near zero if each institution brings its own AI key, if members use the non-AI features and mock mode, or via donated education/nonprofit AI credits. There is no licensing or subscription cost at any point.

Suggested first steps

- A short call to demo a **TWAS-branded instance** so the offering is concrete.
- A **TYAN workshop/webinar** where member-educators build a live course — the best way to seed real usage.
- A light **pilot** with a few member instructors to gather simple evidence TWAS can cite when recommending it more widely.

Contact: Hasan Aljabbouli (NYU), TWAS/TYAN member · Project & source: github.com/hjmacemail/lmsbridge · License: AGPL-3.0